Juwon Lee

Education R&D · Developer, Seoul Robot & AI Museum

Target roles: Education planning & development

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Summary

At the Seoul Robot & AI Museum, I plan education programs and build the software used in classes and exhibits, from student web apps to exhibit kiosks and staff tools. My smart-farm program for the Yangpyeong Education Office was taught as a demo class to 5 high-school students in August 2026, and three instructors I trained taught its 15-student cohort in October. My AI art exhibit generated 14,328 images on the museum floor from June to August 2026.

Experience

Education R&D · Developer · Seoul Robot & AI Museum (via operator FNJ Co., Ltd.) 2026.03 – Present

- Research, build, and run interactive AI education web apps and kiosks that visitors use on the exhibition floor.
- Plan and run education programs, including the smart-farm class the Yangpyeong Education Office commissioned through the museum.
- Planned a 10-session, 20-hour AI and coding course for museum staff (one hour of theory and one hour of individual practice per session; five sessions are AI technology trend studies), running since November 4, 2026.
- Run the Picabot robot-arm drawing studio as a regular 30-minute hands-on AI session for visiting school groups; led its extended 50-minute class at the KT × Seoul City AI Future Camp (100 elementary students attended) in August 2026.
- Build internal tools behind exhibits and classes, including a staff scheduling app used daily by about 20 docents and part-time staff.
- Guide exhibitions in Korean and English and support the multipurpose education rooms.

VQA Dataset Builder & QA Reviewer · Ubase, Inc. (dispatched to Select Star) 2025.11 – 2025.12

- Built and reviewed visual question-answering (VQA) datasets for LLM training on an ETRI (Electronics and Telecommunications Research Institute) project, including complex 3-hop reasoning Q&A sets.
- Checked each question–response pair against the project's correction guidelines for grammatical and semantic accuracy, and rewrote answers that did not follow logically from the question.
- Sustained about 1.5× the team's average throughput while holding accuracy.

Youth Intern · Ministry of Science and ICT 2025.04 – 2025.08

- Supported improvements to the national Pay-TV service quality evaluation, including its user-satisfaction survey, and drafted Broadcasting Act review materials.
- Attended exhibitions, conferences and seminars, and wrote result reports and trend reports.

Selected projects

Smart-Farm Education Program 2026

Seoul Robot & AI Museum × Yangpyeong Education Office · Planning & development, solo · smartfarm-lab.com

IoT smart-farm and AI crop-diagnosis class commissioned by the Yangpyeong Education Office through the museum. Demo class for 5 high-school students in August 2026; in October 2026, three instructors I trained taught it to a 15-student cohort selected by application.

- Built the Next.js web app with server API routes on Vercel and Arduino (C++) firmware for the class's 15 ESP32 student boards (one per student), which write sensor readings straight to Supabase.
- Designed the class flow: a room-picker hub, 4 steps (weather, AI transfer learning, IoT sensors, automation; in the last step the real board switches its LED, fan and pump on thresholds the students set), and a closing quiz.
- Students train a model in the browser: transfer learning with TensorFlow.js on a self-hosted MobileNet v2, with the epoch and loss values of their own run on screen.
- Tied lessons to real growing conditions with a location-based weather API and live sensor values in the lab database.
- The first prototype used a Python/Flask relay and MicroPython firmware.

Tech: TensorFlow.js · MobileNet v2 · Next.js · Supabase · ESP32 · Arduino / C++ · Weather API

Mystery Playground RE:PLAY 2026

Seoul Robot & AI Museum · 4F special exhibition archive · Development, field tuning, operations tooling · On view · replay-exhibit-game.vercel.app

Ten-mission tablet game archiving the museum's 4th-floor special exhibition; five MediaPipe vision models run on the tablet.

- Traced misdrawn avatars to hats and headbands using the first 128 numbers-only visitor logs, then fixed the color, beard and curl rules and checked the change against a frozen eight-image benchmark.
- Mission 6 reduces 478 face landmarks and hair, skin and clothing masks to a numeric feature record with no photo pixels, then draws a full-body pixel-art avatar from it.
- The app installs on a Galaxy Tab as an offline PWA, including about 51 MB of AI runtime and models.
- Operations built in: tablets upload daily mission stats, staff read them on a token-protected page with CSV export, avatar logs expire after 90 days, and English mode returns to Korean after 30 idle seconds.

Tech: JavaScript · MediaPipe · Three.js · PWA · Vercel Functions · Supabase

Raimi's AI Art Lab 2026

Seoul Robot & AI Museum · Planning, development, operations · Ran Jun–Aug 2026

Guided prompt builder where visitors make AI artwork with the museum character Raimi. It ran on the exhibition floor from June to August 2026, and its own backend recorded 14,328 generated images, 9,865 of them in July.

- 10-step (+5 advanced) choice-based prompt builder that shows how a prompt is put together, designed for visitors of all ages.

- Replaced an off-the-shelf GPT service that froze mid-use with a custom web app and serverless backend; no freezes since the switch.
- Serverless backend on the OpenAI image API that watermarks every image with the museum logo, stores images on Cloudflare R2 and generation stats in Supabase, and hands visitors their artwork by QR code.
- Counting rule: one image is one successful generation; blocked or failed requests are not counted.

Tech: JavaScript · OpenAI API · Vercel Serverless · Supabase · Cloudflare R2 · PWA

Exhibit Auto-Recovery 2026

Seoul Robot & AI Museum · self-initiated · Diagnosis, build, rollout, handover · Installed Jul 2026 · still running · approved by the museum

Recovery system for a face- and voice-interaction robot exhibit (MaskBot) whose control software froze or exited mid-operation and could only be restarted by hand from one specific machine.

- Watchdog checks the program every 20 seconds and relaunches it within a minute of a crash or hang it can detect, with a two-stage check against false positives and a 5-minute pause after 3 restarts in a row.
- Recovery page with one button that any staff member can open on their own phone, reached over a Tailscale (WireGuard) mesh VPN with no port open to the internet.
- Silent speech-recognition stalls give no health signal, so twice-daily scheduled restarts and the manual button cover them.
- Wrote a handover document (configuration, logs, routine maintenance, diagnosis playbook, known limits) so the maintenance technician can own the system.

Tech: PowerShell · Windows Task Scheduler · HTTP service · Tailscale / WireGuard

RAIM EN: English Tour Booking 2026

Seoul Robot & AI Museum · Planning, development, deployment · Live since Oct 2026 · approved by the museum

English visit guide and tour-booking site that the museum adopted as its official booking channel for foreign visitors. The city reservation system requires Korean mobile identity verification, so they could not book online; here they book with an email address, and it opened 130 online seats a week at launch where there had been none.

- Built booking around the museum's monthly allocation of sessions and seats: a pending request already holds its seat so sessions cannot be overbooked, November requests wait for staff approval while bookings from December confirm instantly, and cancellations return seats automatically.
- Staff admin with individual accounts, login lockout and CSRF protection, where staff check bookings and mark visits and no-shows.
- Personal data auto-deleted 90 days after the visit (no-show records kept longer); visit statistics store no IP addresses.
- 290+ automated tests (node:test); hosting moved to Fly.io with a persistent volume in October 2026.

Tech: Node.js · Express · SQLite · EJS · Fly.io

Doctor-Green 2024 – 2026

Smart-farm platform · personal project · 2024 prototype: team MAKENEW (PM & developer) · 2026 rebuild: solo

Smart-farm platform that puts IoT environment monitoring and control and crop-disease diagnosis on one web platform. The IoT side was built and verified end to end; the diagnosis model is being rebuilt from the data up.

- ESP32 sensor nodes post directly to Supabase over Wi-Fi with no relay server, a Next.js dashboard polls every 5 seconds, and LEDs and fans follow a desired-state pattern with optimistic UI.
- Built the training set from AI Hub strawberry-disease data: 39,889 crop images in 5 classes (healthy plus 4 diseases) with a group-aware stratified split (train 32,155 / val 4,326 / test 3,408). Classifier training (ConvNeXt-Tiny or EfficientNetV2-S) is next; no results yet.
- Grew out of a 2024 VGG16 plant-disease app built with team MAKENEW (Excellence Award, 2024 ICT·SW Women's Start-up Competition); that model stayed below the team's 95% accuracy bar, which led to the 2026 rebuild.
- Wrote a Windows downloader for the AI Hub dataset that calls its API directly, merges split archives and restores Korean filenames.

Tech: Next.js 16 · TypeScript · Supabase · ESP32 · Python · Vercel

SEOUL RAIM Staff Platform 2026

Seoul Robot & AI Museum · Planning & development · In daily use · approved by the museum

Staff app that brings per-date schedules, the roster, live duty status and notices into one place; about 20 docents and part-time staff use it daily as an internal web app (PWA).

- Rebuilt a single-HTML prototype as a Vite + React codebase with real-time data on Firebase (Firestore).
- Designed the screen flow: role-based login, bottom tabs (home, notices, schedule, status, my page, admin) and a manager PC console.

Tech: React · Vite · Firebase / Firestore · Capacitor

Additional projects

Exhibit Usage Stats Hub (2026 · In operation): One dashboard for the daily usage counts of 9 registered exhibits, with atomic daily counters in Supabase, row-level security with no public policies, and an offline queue for tablet exhibits. Tech: Vercel Functions · Supabase · PostgreSQL · Row Level Security

RAIM 4-Cut Studio (2026 · Seasonal use · ended): Photo kiosk with on-device best-shot suggestions from MediaPipe face scores and QR takeaway; 1,932 uses recorded on 26 days from June 14 to September 6, 2026. Tech: JavaScript · MediaPipe · Canvas · Supabase · PWA

Seeing the Room Through LiDAR (2026 · Installed Sep 2026): Self-touch explainer beside the museum's LiDAR media art that walks visitors through five steps from sensor points to the projected waves; a single page with no runtime libraries for an old TV browser, with a source rule for every sentence on screen. Tech: JavaScript · Canvas 2D · PWA · Blender

Raimi's AI Language Lab (2026 · Ran Jun to Oct 8, 2026): Five touch mini-games on tokenization, embeddings, word order, attention and next-word prediction; a Korean/English offline-capable PWA with no server that ran on 3 exhibition tablets. Tech: JavaScript · HTML/CSS · PWA

I'm a Restorer! (2026 · Development · Once-daily program): Puzzle web game in which visitors restore traditional Korean patterns such as dancheong colors and folk-painting motifs, used on 8 exhibition tablets in a once-daily program of about 15 participants per session. Tech: JavaScript · HTML/CSS · GitHub Pages

AI Persona Companion App (2026 · Deployed Sep 2026): Phone web app for the caricature-robot exhibit, opened from a QR code beside it: in-browser MediaPipe face tracking redraws the face in shapes, and video and face coordinates never leave the phone. Tech: MediaPipe Face Landmarker · JavaScript · Canvas 2D · GitHub Pages

First Time in the Future? Time-Machine AR (2026 · Deployed · on-site launch pending): Tablet AR web app for an education program across 5 spots in the exhibition; it identifies each spot by band color because MindAR could not tell the five posters apart, and was tested with a fake-camera harness in headless Chrome. Tech: MindAR · Three.js · WebGL · PWA

Document Template Converter (2026 · Online · handover pending): Browser-only tool that rebuilds Word, PowerPoint and Hangul (HWP/HWPX) documents in manual templates I designed for the museum and exports Word, PowerPoint or PDF; nothing is uploaded. Tech: JavaScript · mammoth.js · docx · PptxGenJS · JSZip

Teaching, mentoring & activities

2026 SW·AI Donghaeng Project (KOFAC) · Youth mentor · makerthon coaching, live online class mentoring (middle school) 2026.09 – 2026.10

KB LA School (middle school) · Middle-school mentor · twice a week, 80 min · outside work hours 2026.07 – Present

Gwangyang AI Smart i-Kium · Online learning mentor, Grade-3 math & English · once a week · outside work hours 2026.03 – 2026.08

Geuruteogi Learning Mentoring · University mentor (6th cohort) · one-on-one study mentoring for a Grade-9 student · twice a week, 90 min · outside work hours 2026.02 – Present

EBS Online Education Mentoring · Online mentor, elementary Korean language 2025.08 – 2025.12

Youth SW-Donghaeng Hackathon · University mentor 2025.07

CIEE SEOULMATE · Seoulmate (exchange-student buddy) · planned small-group cultural exchange activities, joined monthly CIEE events 2025.06 – 2025.12

SW-Donghaeng Project · Youth mentor · 3 high-school visiting classes over 3 months, coaching students on physical-computing code 2024.09 – 2024.12

Hankyong Start-up Club (MAKENEW) · Team leader · plant-disease diagnosis app 2024.07 – 2025.02

LS Dream Science Class (20th) · Hands-on science experiments for elementary students 2024.06 – 2024.08

KB LA School (high school) · Online mentor · Grade-11 Math I & Chemistry (2024), Grade-10 Math (2025) 2024.03 – 2026.02

Skills & languages

Planning & Education: Product Planning (PM) · Learning Design · Bilingual Exhibition Guiding (KO·EN)
Frontend & App: Next.js · React · TypeScript · JavaScript · Tailwind · PWA · Capacitor
Backend, Infra & Ops: Supabase · Firebase / Firestore · Flask · Serverless Functions · Cloudflare R2 · Vercel · Tailscale / WireGuard
AI & Computer Vision: Python · TensorFlow.js · MobileNet v2 · MediaPipe · VGG16 / CNN · OpenAI API
Data & Datasets: VQA Datasets · Dataset QA · AI Hub · Stratified Splits
Hardware · IoT · Robotics: ESP32 · MicroPython · Arduino / C++ · Sensors
Languages: Korean (native) · English (fluent; TOEIC Speaking AL, OPIc AL)

Awards

Prize · 2025 GH Youth Build-Up Start-up Competition (GH Corp.) 2026.03
Team MAKENEW (PM & developer).

Excellence Award (2nd place, finals) · 2025 Science Museum & Community AI Hackathon (National Science Museum) 2025.09
AI docent concept connecting science museum exhibitions with their local communities (PM & developer).

Excellence Award · ICT·SW Women's Start-up Competition 2024.12
VGG16 plant-disease diagnosis app, the predecessor of Doctor-Green; organized by the Korea IT Businesswomen's Association (PM & developer).

Encouragement Award · 2024 Chungnam Generative AI Technology Start-up Idea Competition 2024.10
The 2024 VGG16 version of Doctor-Green (PM & developer). Hosted by MSIT, NIPA and Chungnam Techno Park; organized by Sun Moon University.

Excellence Award · HK StartUP&GO Audition, Hankyong National University 2024.10
Plant-disease diagnosis app that later grew into Doctor-Green (PM & developer).

Education

Hankyong National University 2022.03 – 2027.02
Plant Life & Environment, double major in Software Convergence · expected to complete graduation requirements (graduation deferred)

Rutgers University 2025.01 – 2025.02
Winter Intensive English Program · New Brunswick, NJ, USA